

Influence of Prophylactic Antibiotics Aiming at Gut Decontamination on Gastrointestinal Graft-Versus-Host Disease and Overall Survival Following Allogeneic Haematopoietic Stem Cell Transplantation

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Abstract

Introduction:

The impact of commensal bacteria harbored in the gastrointestinal tract known as the microbiota has long been recognized as a pivotal factor in allogeneic hematopoietic stem cell transplantation (aHSCT). The microbiota is at the origin of acute graft-versus-host disease (aGVHD) and infections, two important lethal complications in aHSCT. High-dose conditioning chemotherapy prior aHSCT disrupts the gut epithelial barrier allowing bacterial by-products to translocate into the peripheral blood and modulates T cell response and pro-inflammatory cytokines. Such observation led to an effort by certain transplant

centers to eliminate bacterial colonization prior to aHSCT to decrease the risk of gram-negative bacterial translocation. In this study, we assessed whether patients' gut decontamination prior to allogeneic stem cell infusion influenced the frequency of aGVHD, pneumomatoses coli (PC) a significant complication of gastrointestinal (GI) GVHD and overall survival.

Methods:

We retrospectively reviewed the charts of 543 patients who had undergone a single myeloablative or nonmyeloablative aHSCT for hematological malignancies from two academic hospitals in the province of Quebec, Canada between January 2005 and December 2012. Exclusion criteria included prior aHSCT, syngeneic and haploidentical aHSCT. Each university hospital has implemented a different pre-transplant antibioprophylaxis guideline. At HMR hospital, ciprofloxacin or moxifloxacin were started at initiation of the conditioning regimen for gut decontamination, but were omitted in patients with fluoroquinolone or penicillin allergy and during nosocomial infections outbreak. On the other hand, in the CHU de Quebec patients were not prescribed prophylactic antibiotics (ATB). In addition, ATB used to treat infections before the stem cells infusion were considered. To determine the impact of ATB, we performed multivariable analyses adjusting for the following confounding factors: age, gender, stem cell origin (source), donor type/match, and conditioning regimens to compare the frequency of aGVHD, PC, leucocytes recovery at day+14 and overall survival (OS) in patients receiving or not ATB.

Results:

500 patients were included and a total of 240 (48%) patients received ATB at the time of conditioning regimen. Demographics were similar in both groups with mean age of 48 years. Frequency of grade II-IV aGVHD was more elevated in patients receiving ATB compared to the no ATB group (42% vs 28% respectively with adjusted OR (aOR)=1.53 ($p<0.05$). The severity of the aGVHD in the ATB group was driven by the GI-GVHD with a higher level of grade II-IV (20.7%) compared to no ATB group (10.8%) with aOR=2.00 ($p<0.01$). Severity of skin and liver GVHD were similar in both groups. Among the 12 patients that developed PC diagnosed on CT-scan, all received ciprofloxacin during conditioning and PC was associated with 80% mortality. The difference of aGVH frequency may have translated into survival as the ATB group was associated to lower 1 year survival compared to no ATB group (74% vs 88%, OR=0.36 and aOR=0.38 ($p<0.05$). This significant survival difference at 1 year persisted over time and median OS was 4 and 5 years respectively ($p<0.05$). Taking into consideration the entire follow-up period of 10 years, the hazard rates associated with ATB were estimated at 1.61($p<0.06$) and 1.43 ($p<0.05$) after adjusting for clinical parameters. Interestingly, post-hoc analysis revealed independent impact of the ATB on D+14 neutrophils. This was reflected by a lower neutrophil count in patients on ATB that received stem cells from a match-related donor compared to no ATB counterpart with a ratio of 0.38 ($p<0.05$).

Discussion:

This retrospective study indicated that ATB were associated to a more severe aGVHD driven by digestive manifestations of the GVHD and higher incidence of PC even after multivariable analysis. These life-threatening complications may impact on the 1-year and OS that were lower in the ATB group. Without undermining the role of ATB prophylaxis to prevent infection in aHSCT setting, treatment with ATB impact on the commensal microbiota and diminishes its diversity. This imbalance created by the ATB may have contributed to the pathophysiology of GI-GVHD. This ultimately highlights the importance to reconsider antibioprophylaxis to preserve an intact flora and its benefits in aHSCT.

Disclosures

No relevant conflicts of interest to declare.

Author notes

*Asterisk with author names denotes non-ASH members.